The 3-adic mixing rate of the Syracuse random variable: reductions, a boundary-layer mechanism, and a sharp-rate conjecture

Hikaru Wakaura*

October 16, 2026

Abstract

Let δ_m be the maximal nontrivial Fourier coefficient of the Syracuse random variable modulo 3^m — the quantity whose decay drives Tao's almost-all result for the Collatz map; Tao proved superpolynomial decay and flagged exponential decay as open. We prove five structural theorems: an exact one-dimensionalization of the Fourier problem onto the exponent lattice; a first-passage renormalization identity; a weighted contraction lemma with optimal rate exactly 2^{-I_0}, $I_0 = (1 - H(\log_3 2))/\log_3 2$; the exact Kesten tail exponent 1; and an exact geometric law for the 2-adic valuations of $2^K \bmod 3^m$ on full periods. With a window-audited weighted-ℓ^1 bookkeeping these yield a proof architecture for $\delta_m \le \mathrm{poly}(m)\, 2^{-rm}$, $r > 0.045$, with two inputs: a local equidistribution hypothesis (a finite-orbit quantitative form of Furstenberg $\times 2 \times 3$ rigidity) and band-rate estimates decidable on finite frequency sets, which we certify with error accounted on the full window at $m = 64, 128, 256$. Certified computations to $m = 1200$ support the sharp-rate conjecture $\delta_m = \Theta(2^{-I_0 m} m^{-c})$.

Keywords: $3x + 1$ problem; Syracuse random variable; Fourier decay; 3-adic equidistribution; Kesten chain; Furstenberg $\times 2 \times 3$ rigidity; experimental mathematics.
2020 Mathematics Subject Classification: Primary 11B83; Secondary 11K16, 37P10, 60H25.
Word count: approximately 5,000 words.

1 Introduction

The Collatz $(3x + 1)$ conjecture concerns the map on positive integers that sends even n to $n/2$ and odd n to $3n + 1$. The strongest known result, due to Tao [1], is that almost all orbits (in logarithmic density) attain almost bounded values. At the analytic heart of that proof sits a single quantitative object: the 3-adic equidistribution of the *Syracuse random variable*, the random 3-adic integer

$$\mathbf{Syrac} \;=\; \sum_{j\ge 1} 3^{j-1} 2^{-A_j}, \qquad A_j = a_1 + \cdots + a_j, \quad a_i \text{ i.i.d., } \mathbb{P}(a_i = a) = 2^{-a},$$

*QuantScape Inc., 4-11-18, Manshon-Shimizudai, Meguro, Tokyo, 153-0064, Japan. Corresponding author; email: hikaruwakaura@gmail.com.

which describes the residues mod 3^m of the accelerated Collatz iteration started from a random odd integer. Writing $\mathrm{e}(x) = e^{2\pi i x}$, the quantity that controls everything is the maximal nontrivial Fourier coefficient

$$\delta_m \;=\; \max_{\xi \in \mathbb{Z}/3^m\mathbb{Z},\ 3\nmid \xi} \bigl|\mathbb{E}\,\mathrm{e}(\xi\,\mathbf{Syrac}/3^m)\bigr|.$$

Tao's Proposition 1.17 proves $\delta_m \ll_A m^{-A}$ for every fixed A — superpolynomial decay — and Section 1.4 of [1] states explicitly that an improvement to $O(\exp(-cm))$ may be possible but is not needed there and not pursued. A survey of the citing literature (Section 9) shows that no later work supplies the exponential improvement. Its value would be more than aesthetic: the rate of δ_m is the rate-limiting constant in any *effective* version of Tao's theorem.

However, the exact decay rate of δ_m — even its numerical value — appears never to have been determined. Here we determine it numerically with certified lattice computations, identify its mechanism, prove the structural theorems that reduce the upper-bound problem to one isolated arithmetic hypothesis plus finitely decidable band-rate inputs, and certify the latter — with all numerical error accounted — on the full frequency window at fixed levels $m \le 256$. Our results are of four kinds, and we keep the four statuses — *theorem*, *computational fact* (exact or certified computation), *conditional proposition*, and *conjecture* — explicitly labeled throughout.

Theorems

(i) Since 2 is a primitive root mod 3^m, every nontrivial frequency is a power of 2, and the two-variable Fourier problem collapses, with no assumptions, to a one-dimensional recursion over the exponent lattice, with δ_m non-increasing in m and the underlying residue chain exactly stationary after m steps (Theorem 2.1). (ii) A first-passage renormalization identity expresses g_m in terms of shorter chains, with the certified kernel bound $|F(d,r)| \le 2^{-r}\delta_{d-1}$ and an exact identification of the phase-free region with the characteristic function of the real Syracuse variable (Theorem 3.1). (iii) A weighted contraction lemma: with weight $2^{-\lambda u}$ in the coordinate $u = k - m\log_2 3$, the phase-free weighted row sum equals $f(\lambda) = 3^\lambda/(2^{1+\lambda}-1)$, and

$$\min_\lambda f(\lambda) \;=\; 2^{-I_0}, \qquad I_0 = \frac{1-H(\theta)}{\theta} = D\bigl(\mathrm{Geom}(\theta)\,\|\,\mathrm{Geom}(1/2)\bigr) = 0.0793186\ldots,$$

$\theta = \log_3 2$, H the binary entropy (Theorem 4.1) — below the critical line no phase cancellation is needed to contract at exactly the conjectured rate, so the upper-bound problem reduces to the near-line and above-line frequencies. (iv) The real Syracuse variable is a Kesten chain whose Kesten equation $\mathbb{E}[(3\cdot 2^{-a})^\kappa] = 1$ holds at $\kappa = 1$ *exactly* (the identity $3 = 2^2 - 1$), placing it at the Cauchy-type boundary, $\mathbb{P}(Y > t) \sim C/t$ with $C = 1/\log(27/16)$ (Theorem 5.1). (v) An exact 2-adic law on full periods: the identity $\nu_2(r(K)) \ge \sigma \iff |r(K-\sigma)| \le M/2^{\sigma+1}$ for signed residues $r(K)$ of 2^K mod $M = 3^m$ converts 2-adic valuation into Archimedean size, and yields that within any symmetric ball the valuation law over a full period is geometric with explicit error (Theorem 6.2).

Conditional proposition

Assembling (i)–(v) with a capped weighted-ℓ^1 bookkeeping on the frequency window (audited exactly on all 399 one-step inequalities between consecutive computed levels, with zero violations) and sharp product bounds along Tao's two-dimensional update process, we obtain a proof architecture for

$$\delta_m \;\le\; \mathrm{poly}(m)\cdot 2^{-r\,m}, \qquad r > 0.045,$$

— the qualitative form of the improvement flagged in [1] — with two unproven inputs. One is a set of band-rate estimates (BR) for the product bounds, which are decidable on any finite set of frequencies and which we certify on the full extended window for $m \le 256$ (Section 10); the other (Hypothesis 9.4) is a local equidistribution statement: entries of the orbit $\{2^e \bmod 3^n\}$ into a short symmetric interval have geometrically decaying 2-adic valuations, uniformly over windows of length $O(\log q)$, $q = 3^n$. We show (Section 9) that this statement is a finite-orbit quantitative form of Furstenberg's $\times 2 \times 3$ phenomenon [9], that the effective results of Bourgain–Lindenstrauss–Michel–Venkatesh [10] do not reach it, and that the window length $O(\log q)$ is exponentially below the reach of the known exponential-sum technology; it is, however, decidable in time $O(m^4)$ for each fixed m.

Computational facts

Certified exact-arithmetic computations give: full-frequency scans for $m \le 15$; rescaled window computations to $m = 1200$ showing that the extremal frequency exponent satisfies $k^* = m \log_2 3 - (7.0 \pm 0.3)$ with the deficit *constant* over $m = 200$–1200, and a decay rate in $[0.0790, 0.0800]$ bits per level consistent with $I_0 = 0.07932$; a boundary-layer structure theory (Section 8); and certified instances on the full extended frequency window at $m = 64$, 128, 256, 320: unconditionally $\sup_{k \in W_m} |g_m(2^k)| \le 2^{-4.97}$, $2^{-8.4}$, $2^{-15.0}$, $2^{-18.4}$ respectively, and the band-rate inputs (BR) hold on W_m outside the critical strip with positive slack at rates 0.0350, 0.0406, 0.0457, 0.0481 — a rate that rises slowly with m, the product-bound rates near the window edge extrapolating to ≈ 0.049 (Section 10). Statements about δ_m itself at fixed m additionally require the extremal frequency to lie in W_m, which is verified exhaustively for $m \le 15$.

Conjecture

The sharp rate (Conjecture 11.1): $\delta_m = \Theta(2^{-I_0 m}\, m^{-c})$. Equivalently, the 3-adic mixing bottleneck of Syracuse is the cycle-resonance line $2^A \approx 3^k$, and the mixing rate coincides, after clock conversion, with the classical stopping-time density exponent $1 - H(\theta)$: one constant read in three clocks.

Method

This paper practices experimental mathematics with pre-registered decision criteria: for each major numerical discrimination we declared acceptance/rejection thresholds before computing, and we report the outcomes — including a rejected candidate rate — in Appendix A, which doubles as a calibration log. All certified quantities use exact integer or interval-safe arithmetic; scripts and frozen data accompany the paper. A companion paper [15] studies the 2-adic (certificate-complexity) side of the same constant.

2 Setup and the one-dimensional reduction

Fix $m \ge 1$ and let w_0 be arbitrary in $\mathbb{Z}/3^m\mathbb{Z}$. The Syracuse residue chain is

$$w_t \;=\; (3w_{t-1} + 1) \cdot 2^{-a_t} \bmod 3^m, \qquad a_t \text{ i.i.d., } \mathbb{P}(a_t = a) = 2^{-a}\ (a \ge 1), \tag{1}$$

where 2^{-1} denotes the inverse of 2 mod 3^m. This is the law of the residues mod 3^m of the accelerated Collatz map $n \mapsto (3n+1)/2^{\nu_2(3n+1)}$ along the odd iterates, in the sense made precise in [1, §1]; the stationary law of (1) is the law of **Syrac** mod 3^m.

Theorem 2.1 (one-dimensional reduction)**.** *(a) 2 is a primitive root mod 3^m for every $m \geq 1$:* $\operatorname{ord}(2 \bmod 3^m) = 2 \cdot 3^{m-1} = |(\mathbb{Z}/3^m\mathbb{Z})^\times|$.

(b) Consequently every ξ with $3 \nmid \xi$ is a power of 2 mod 3^m, and with $g_m(k) := \mathbb{E}\, \mathrm{e}\big(2^k \mathbf{Syrac}/3^m\big)$ (exponents read mod ord(2)*) one has $\delta_m = \max_k |g_m(k)|$ and the exact lattice recursion*

$$g_m(k) \;=\; \sum_{a \geq 1} 2^{-a}\, \mathrm{e}\Big(\tfrac{2^{k-a} \bmod 3^m}{3^m}\Big)\, g_{m-1}(k-a). \tag{2}$$

(c) The chain (1) is exactly *stationary after m steps: for $t \geq m$ the law of w_t equals the law of* **Syrac** *mod 3^m, for every initial w_0.*

(d) $\delta_m \leq \delta_{m-1}$ for all $m \geq 2$.

Proof. (a) From $2^2 = 1 + 3$ and the lifting-the-exponent identity $(1 + 3^s u)^3 \equiv 1 + 3^{s+1}u \pmod{3^{s+2}}$ one gets $\nu_3(2^{2 \cdot 3^t} - 1) = t + 1$ exactly, so the order of 2 mod 3^m divides $2 \cdot 3^{m-1}$ and is divisible by 3^{m-1}; and $2^{3^j} \equiv (-1)^{3^j} = -1 \not\equiv 1 \pmod 3$ shows the order is even. Hence $\operatorname{ord}(2) = 2 \cdot 3^{m-1}$.

(b) By (a) the cyclic group generated by 2 is the whole unit group, so the powers of 2 exhaust all ξ with $3 \nmid \xi$, and the maximum defining δ_m is attained on the exponent lattice. The recursion follows by conditioning (1) on the last step: with $\xi = 2^k$,

$$\mathrm{e}\big(\xi(3w+1)2^{-a}/3^m\big) = \mathrm{e}\big(2^{k-a} \cdot 3w/3^m\big)\, \mathrm{e}\big(2^{k-a}/3^m\big) = \mathrm{e}\big(2^{k-a}w/3^{m-1}\big)\, \mathrm{e}\big(2^{k-a}/3^m\big),$$

and taking expectations over $w \sim \mathbf{Syrac} \bmod 3^{m-1}$ and a gives (2).

(c) By induction on j: $w_t \bmod 3^j$ depends only on $(a_{t-j+1}, \ldots, a_t)$, because $(3w+1) \bmod 3^j$ depends only on $w \bmod 3^{j-1}$ and multiplication by the unit 2^{-a} preserves this. For $j = m$, $t \geq m$, the law of w_t is a fixed function of the last m steps, independent of w_0; it is therefore the unique stationary law.

(d) Bound (2) by the triangle inequality: $|g_m(k)| \leq \sum_a 2^{-a}|g_{m-1}(k-a)| \leq \delta_{m-1}$, using that every frequency 2^{k-a} is again a unit mod 3^{m-1}. □

Remark 2.2. Theorem 2.1(b) upgrades to a theorem what would otherwise be a heuristic restriction to special frequencies: *every* nontrivial frequency lives on the exponent lattice, so the recursion (2) loses nothing. Full-frequency exhaustive scans for $m \leq 15$ confirm that the maximizer is always a power of 2 in a narrow window of exponents (Section 7).

3 First-passage renormalization

Let $Y_m = \sum_{j=1}^m 3^{j-1} 2^{-A_j}$ denote the *real* Syracuse variable (the same series read in $\mathbb{R}$), and write $x = 2^k/3^m$.

Theorem 3.1 (renormalization identity)**.** *For all m, k:*

(a) (phase-free region) On the event $\{A_m \leq k\}$ the 3-adic phase coincides exactly with the real one: $\mathrm{e}(2^k\, \mathbf{Syrac}_m/3^m) = \mathrm{e}(2^k Y_m/3^m)$ *pathwise, where $\mathbf{Syrac}_m$ is the level-m truncation. Hence* $\mathbb{E}[\mathrm{e}(2^k \mathbf{Syrac}_m/3^m); A_m \leq k] = \mathbb{E}[\mathrm{e}(xY_m); A_m \leq k]$.

(b) (first-passage decomposition) With $\mathrm{amp}_j(A) = \mathbb{E}[\mathrm{e}(xY_j); A_j = A]$ *and*

$$F(d,r) \;=\; \sum_{a>r} 2^{-a}\,\mathrm{e}\Big(\tfrac{2^{\,r-a} \bmod 3^d}{3^d}\Big)\,V_{d-1}(2^{\,r-a}), \qquad V_{d-1}(\xi) = \mathbb{E}\,\mathrm{e}(\xi\,\mathbf{Syrac}_{d-1}/3^{d-1}),$$

one has the exact identity

$$g_m(k) \;=\; \mathbb{E}[\mathrm{e}(xY_m); A_m \le k] \;+\; \sum_{j<m}\;\sum_{A\le k} \mathrm{amp}_j(A)\,F(m-j,\,k-A),$$

obtained by splitting at the first step where the valuation sum exceeds k*.*

(c) (kernel bound) $|F(d,r)| \le 2^{-r}\,\delta_{d-1}$ *for* $d \ge 2$*.*

Proof. (a) On $\{A_m \le k\}$ every exponent $k - A_j$ ($j \le m$) is non-negative, so $2^k \cdot (2^{-A_j} \bmod 3^m) \equiv 2^{k-A_j} \bmod 3^m$ with 2^{k-A_j} a genuine integer, and $\mathrm{e}\big(\sum_j 3^{j-1}2^{k-A_j}/3^m\big)$ is simultaneously the 3-adic and the real phase.

(b) Decompose according to the first index $j+1$ with $A_{j+1} > k$ (if none, we are in the event of (a)). Conditioning on $A_j = A \le k$ and on the overshooting step $a_{j+1} = a > k - A$, the remaining $m - j - 1$ steps form a fresh chain at level $m - j - 1$, and collecting phases gives the stated kernel with $r = k - A$; the identity is exact. (It was additionally verified numerically to relative error 10^{-14} across levels.)

(c) $|F(d,r)| \le \sum_{a>r} 2^{-a}\,|V_{d-1}(2^{r-a})| \le \delta_{d-1}\sum_{a>r} 2^{-a} = 2^{-r}\delta_{d-1}$, again because the frequencies are units. □

Remark 3.2 (why absolute values do not suffice)**.** It is tempting to iterate Theorem 3.1(b)–(c) with absolute values throughout. This provably fails to close: the phase-free main term must then be bounded by the large-deviation probability $\mathbb{P}(A_m \le k)$, and near the critical line the inductive inequality loses a constant factor per level that destroys any exponent. The failure is structural — on the good event the final phase still rotates by at least $1/3$ of a turn, so positivity-based lower bounds are equally impossible, and the modulus of g_m is carried by cancellation among crossing terms (Section 8). This motivates the weighted approach of the next section.

4 Weighted contraction and the reduction of the upper bound

Write $\alpha = \log_2 3$ and use the line coordinate $u = k - m\alpha$.

Theorem 4.1 (weighted contraction)**.** *For* $\lambda > -1$ *define the phase-free weighted row sum of the recursion* (2) *with weight* $2^{-\lambda u}$ *(so that* $\lambda^* > 0$ *below corresponds to a weight decaying above the line, matching the convention* $\mu = -\lambda^*$ *of* (4)*):*

$$f(\lambda) \;=\; \sum_{a\ge 1} 2^{-a}\,2^{\lambda(\alpha-a)} \;=\; \frac{3^\lambda}{2^{1+\lambda}-1}.$$

Then f *is strictly log-convex on* $(-1,\infty)$ *and*

$$\min_\lambda f(\lambda) \;=\; 2^{-I_0}, \qquad I_0 = \frac{1-H(\theta)}{\theta}, \qquad \text{attained at } 2^{1+\lambda^*} = \frac{\alpha}{\alpha-1}, \quad \lambda^* = 0.4380326\ldots$$

Applying the weight to (2) and bounding phases by 1 wherever the weight decays, one obtains for the weighted norm $\|g_m\|_\lambda = \sup_k 2^{-\lambda u}|g_m(k)|$ the one-step inequality $\|g_m\|_\lambda \le f(\lambda)\|g_{m-1}\|_\lambda +$ (boundary terms supported near and above $u = 0$): with no phase input at all, the below-line region contracts at exactly the conjectured rate, and the upper-bound problem for δ_m reduces to the control of frequencies with $u \gtrsim 0$. The precise audited form of this reduction is the capped functional described after the proof.

Proof. The row sum is a geometric series: $\sum_a 2^{-a(1+\lambda)} = 1/(2^{1+\lambda} - 1)$, times $2^{\lambda\alpha} = 3^\lambda$. For the minimum, substitute $t = 2^{1+\lambda} > 1/2$: then $3^\lambda = t^\alpha/3$ and $f = t^\alpha/(3(t-1))$, with critical point $\alpha(t-1) = t$, i.e. $t^* = \alpha/(\alpha-1)$, and

$$\min f = \frac{(t^*)^\alpha}{3(t^*-1)} = \frac{\alpha^\alpha}{3\,(\alpha-1)^{\alpha-1}}.$$

Taking $-\log_2$ and using $\log_2 3 = \alpha$: $-\log_2 \min f = \alpha - \alpha\log_2\alpha + (\alpha-1)\log_2(\alpha-1)$. On the other hand, with $\theta = 1/\alpha$,

$$\frac{1-H(\theta)}{\theta} = \alpha\Big(1 - \tfrac{1}{\alpha}\log_2\alpha - \big(1-\tfrac{1}{\alpha}\big)\log_2\tfrac{\alpha}{\alpha-1}\Big) = \alpha - \alpha\log_2\alpha + (\alpha-1)\log_2(\alpha-1),$$

which is the same expression. (The identity was additionally checked to 50 digits.) Log-convexity holds because $f(\lambda)$ is the moment generating function $\mathbb{E}\,2^{\lambda(\alpha-a)}$ of the step variable. □

Two consequences organize everything that follows. First, I_0 is simultaneously a Kullback–Leibler divergence:

$$I_0 \;=\; D\big(\mathrm{Geom}(\theta)\,\|\,\mathrm{Geom}(1/2)\big), \qquad \theta\, I_0 = 1 - H(\theta), \tag{3}$$

the second equality being a one-line identity; the tilted step law at the optimum is exactly $\mathrm{Geom}(\theta)$, whose mean step is $1/\theta = \alpha$ — the drift that tracks the critical line. Second, the reduction can be run with a *free* weight exponent. For $\mu < 0$ define

$$r(\mu) \;=\; \mu\,\alpha + \log_2\big(2^{\,1-\mu} - 1\big), \tag{4}$$

the contraction rate of the capped weighted-ℓ^1 *window* functional

$$N_\mu(m) = \sum_{k\in W_m} \min\big(2^{\mu u},\,1\big)\,|g_m(k)|,$$

W_m the extended window of Section 10; one checks $\max_\mu r(\mu) = I_0$ at $\mu = -\lambda^*$, and smaller $|\mu|$ trades rate for weaker demands on the frequencies below the window. The index set must be a window: summed over all of $\mathbb{Z}$ the functional diverges by periodicity of g_m, and on a centered fundamental domain of the exponent circle it does not contract (the two levels have different periods), so the global formulation carries no content beyond the window one; frequencies entering the window from below are the band-rate inputs (BR) of Section 9.

Computational Fact 4.2 (audited bookkeeping)**.** The one-step inequality $N_\mu(m) \le 2^{-r(\mu)}N_\mu(m-1)$ of the window functional with $\mu = -0.16$, $r(\mu) = 0.0504$, was audited against the exact lattice data for all 399 pairs of consecutive levels $2 \le m \le 400$: zero violations, with no injection terms required within the window. The bookkeeping at the window boundary is exactly the constraint system (BR) used in the certified instances of Section 10.

5 Kesten structure of the real variable

Theorem 5.1. *The real Syracuse variable* $Y = \sum_{j\ge 1} 3^{j-1}2^{-A_j}$ *satisfies the stochastic fixed-point equation* $Y \overset{d}{=} 3\cdot 2^{-a}\,Y + 2^{-a}$ *(*$a \sim \mathrm{Geom}(1/2)$ *independent of* Y *on the right), and its Kesten equation* $\mathbb{E}[(3\cdot 2^{-a})^{\kappa}] = 1$ *holds* exactly *at* $\kappa = 1$*:*

$$\mathbb{E}\big[3\cdot 2^{-a}\big] = \sum_{a\ge 1} 2^{-a}\cdot 3\cdot 2^{-a} = 3\sum_{a\ge 1} 4^{-a} = 3\cdot\tfrac{1}{3} = 1.$$

The multiplier $\log(3\cdot 2^{-a}) = \log 3 - a\log 2$ *takes its values in the shifted lattice* $\log 3 - (\log 2)\mathbb{Z}$*, whose generated closed subgroup is dense because* $\log 3/\log 2$ *is irrational; its law is therefore* non-arithmetic*, the Goldie conditions hold, and*

$$\mathbb{P}(Y > t) \sim \frac{C}{t}, \qquad C = \frac{\mathbb{E}[2^{-a}]}{\mathbb{E}[3\cdot 2^{-a}\log(3\cdot 2^{-a})]} = \frac{1}{\log(27/16)} = 1.91114\ldots,$$

so $\mathbb{E}Y = \infty$ *and the tail exponent is exactly* 1*.*

Proof. The fixed-point equation restates the definition after one step. The Kesten computation at $\kappa = 1$ is the displayed line — $3^{\kappa} = 2^{\kappa+1} - 1$ has the integer solution $\kappa = 1$ because $3 = 2^2 - 1$. The moment condition of implicit renewal theory [7, 8] is immediate ($\mathbb{E}[(3\cdot 2^{-a})\log^{+}(3\cdot 2^{-a})] < \infty$), and in the non-arithmetic case Goldie's theorem gives $\mathbb{P}(Y > t) \sim Ct^{-1}$ with $C = \mathbb{E}[B]/\mathbb{E}[A\log A]$ for $A = 3\cdot 2^{-a}$, $B = 2^{-a}$; here $\mathbb{E}[B] = 1/3$ and $\mathbb{E}[A\log A] = \log 3 - \frac{4}{3}\log 2 = \frac{1}{3}\log(27/16)$. □

This places Syracuse at the *boundary* Cauchy-type case: power corrections in $1/m$ are to be expected, which is exactly what the numerics of Section 7 display, and the polynomial factor in Conjecture 11.1 should be regarded as this mechanism's signature rather than as numerical noise.

6 An exact 2-adic law on full periods

The conditional route of Section 9 needs one probabilistic input about the deterministic orbit $\{2^K \bmod 3^m\}$: entries into a short symmetric interval should have geometrically distributed 2-adic valuations. On *full periods* this is a theorem, by an elementary identity that converts the 2-adic condition into an Archimedean one.

For odd $M \ge 3$ let $r(K) \in (-M/2, M/2]$ denote the signed residue of 2^K mod M.

Lemma 6.1 (valuation–size identity)**.** *For all* K *and* $\sigma \ge 0$*:*

$$\nu_2\big(r(K)\big) \;\ge\; \sigma \qquad\Longleftrightarrow\qquad \big|r(K-\sigma)\big| \;\le\; \frac{M}{2^{\sigma+1}}.$$

Proof. ($\Rightarrow$) If $2^{\sigma} \mid r(K)$ then $r(K)/2^{\sigma}$ is an integer congruent to $2^{K-\sigma}$ mod M with absolute value at most $M/2^{\sigma+1} < M/2$, hence it *is* the signed residue $r(K-\sigma)$. ($\Leftarrow$) If $|r(K-\sigma)| \le M/2^{\sigma+1}$ then $2^{\sigma}r(K-\sigma)$ is congruent to 2^K mod M with absolute value at most $M/2$, hence equals $r(K)$, which is thus divisible by 2^{σ}. □

Theorem 6.2 (geometric law on cycles)**.** *Let $M = 3^m$ and let K range over a full period of the orbit $\{2^K \bmod 3^m\}$ (length $2 \cdot 3^{m-1}$, all units, by Theorem 2.1(a)). Then for every $\varepsilon \in (0, 1/2]$ and every $\sigma \geq 0$,*

$$\mathbb{P}_K\Big(\nu_2(r(K)) \geq \sigma \;\Big|\; |r(K)| \leq \varepsilon 3^m\Big) \;=\; 2^{-\sigma} \;+\; O\Big(\frac{2^\sigma}{\varepsilon\, 3^m}\Big),$$

i.e. inside any symmetric ball the valuation law over a full period is geometric of ratio $1/2$, with constant $C = 1$, up to the stated counting error.

Proof. By Lemma 6.1 (applied once more to intersect with the ball), $\{\nu_2(r(K)) \geq \sigma,\ |r(K)| \leq \varepsilon 3^m\}$ is, under $K \mapsto K - \sigma$, in bijection with $\{|r(K')| \leq \varepsilon 3^m/2^\sigma\}$. Since 2 is a primitive root, over a full period the map $K \mapsto 2^K$ visits every unit exactly once, so the count of K' with $|r(K')| \leq x$ equals the number of nonzero integers prime to 3 in $[-x, x]$, which is $\frac{4}{3}x + O(1)$. Taking the ratio of the counts at $x = \varepsilon 3^m/2^\sigma$ and $x = \varepsilon 3^m$ gives $2^{-\sigma}(1 + O(2^\sigma/\varepsilon 3^m))$. □

Remark 6.3. An independent derivation via Palm sampling of ball entries and a conductance comparison gives the same conclusion without the bijection. Exhaustive verification for $m \leq 12$ over the full period and $\sigma \leq 6$ found zero mismatches of Lemma 6.1, and the observed frequencies at $m = 12$, 0.5000, 0.2500, 0.1250, 0.0625 (for both $\varepsilon = 1/2$ and $\varepsilon = 1/10$), match the geometric law. We also note a caveat used below: value chains generated by non-units (multiples of 9) live on leaves of effective modulus 3^{m-2j+2}, and Lemma 6.1 is applied at the effective scale.

7 Certified numerics to $m = 1200$

All computations use the recursion (2) on the exponent lattice, in exact or error-tracked arithmetic; scripts and data accompany the paper.

Computational Fact 7.1 (extremal frequency and rate)**.** (a) Full-frequency scans for $m \leq 15$ (all $2 \cdot 3^{m-1}$ frequencies): the maximizer of $|g_m|$ is always a power of 2 with exponent in a narrow window at the critical line; the window computation reproduces the true δ_m to ratio 1.0000.

(b) Rescaled lattice computations for $m \leq 1200$: the extremal exponent satisfies $k^* = m \log_2 3 - (7.0 \pm 0.3)$, with the deficit constant over $m = 200$–1200; the local decay rate of the window maximum lies in [0.0790, 0.0800] bits/level, straddling $I_0 = 0.07932$ within the measurement resolution, consistent with a polynomial correction.

(c) Only one side of (b) is accessible to the theory: weighted contraction (Theorem 4.1) can only prove rate $\geq I_0 - \epsilon$ statements for the certified functionals, and the tilted optimum behaves as Geom(θ).

(d) The above-line profile is monotone and steep: the sup over $u \in (0, 2]$ is 4.6% of the ridge value, 9×10^{-5} over $u \in (10, 20]$, 4×10^{-13} over $u \in (40, 80]$ (computed in a truncation-free extended window at $m = 100, 200$): there are no deep resonances above the line.

The caveat that a window computation might miss a distant frequency among the $\sim 10^{286}$ at $m = 600$ is answered structurally: the ℓ^4 functional $S_4'(m) = \sum_{3 \nmid \xi} |\mathbb{E}e(\xi X/3^m)|^4$ is a *computable global scalar* — an FFT sees every frequency — with the collision representation (with $Z = X_1 + X_2 - X_3 - X_4$ for four independent copies) $S_4'(m) = 3^m\, \mathbb{P}(Z \equiv 0 \bmod 3^m) - 3^{m-1}\, \mathbb{P}(Z \equiv$

$0 \bmod 3^{m-1}$), the subtracted term removing the frequencies divisible by 3; exact computation for $m \le 13$ shows $Q(m) = S_4'/\delta_m^4$ saturating at ≈ 18, so the fourth moment is concentrated on the extremal ridge and $\delta_m = (S_4'/Q)^{1/4}$. The four-fold weighted row sum factorizes exactly, $\min_\lambda[3^{-\lambda}/(2^{1-\lambda}-1)]^4 = 2^{-4I_0}$ (the joint four-walk representation is essential: Jensen on the marginal yields only 2^{-I_0}), and the windowed ℓ^4 per-level rate converges one-sidedly to $4I_0 = 0.31727$ ($0.403 \to 0.332$ at $m = 400$).

8 The boundary-layer mechanism

Where does the extremal amplitude live? The following structure statements are certified computations (with pre-registered criteria; outcomes, including one rejected hypothesis, in Appendix A).

Computational Fact 8.1 (boundary layer)**.** (a) (main-term failure) The phase-free main term $\mathbb{E}[\mathrm{e}(xY_m); A_m \le k^*]$ decays at rate $I_0 + 0.0093$ — strictly faster than δ_m; positivity of the good event is destroyed exponentially, and crossing terms carry 80–257 times the main term. Any proof of the lower bound must be an oscillatory one.

(b) (corner localization) In the first-passage decomposition of Theorem 3.1, 70% of the contribution mass to $g_m(k^*)$ sits in 9–18 cells around the corner $r = k^* - A = 0$, $d = m - j = 5$, with d *exactly constant* over $m = 24$–100, and geometric decay $2^{-1.5r}$ in r: the extremal amplitude is generated by paths whose valuation sum first reaches the critical value $O(1)$ levels before the end — a single pinned boundary-layer bridge.

(c) (pinned bridge attains the rate) The single pinned amplitude obeys $|\mathrm{amp}_{m-5}(k^*)| = C\,2^{-am}m^{-1.59}$ with $a = 0.07914 \in [I_0 - 0.001, I_0 + 0.002]$ for $m \le 400$: no exponential phase cancellation occurs in the bridge. Phase retention decays polynomially, $\sim m^{-1.06}$, distributed across scales: the octave factors of the retention profile $K(s, m)$ (which is m-independent for $s \ll m$) lie in $[0.62, 0.95]$ for $s \in [4, 256]$ at $m = 640$ — uniformly bounded away from 0, with no catastrophic octave.

(d) (decorrelation length) The autocorrelation of g_m along the exponent lattice has universal profile with correlation length ≈ 8 exponent units (matching the ridge width and the deficit ≈ 7), measured identically at $m = 200$ and 400; this is the mechanism separating the joint rate $4I_0$ from the marginal I_0 in the ℓ^4 functional.

Summarizing the mechanism: δ_m is a boundary-layer phenomenon on the cycle-resonance line $2^A \approx 3^k$ — the same Diophantine line that governs the nontrivial-cycle problem — with amplitude carried by a single pinned crossing and only polynomial phase loss. The two remaining asymptotic lemmas are stated in Section 11.

9 Toward unconditional exponential decay

9.1 Literature status

Proposition 1.17 of [1] gives $\delta_m \ll_A m^{-A}$, uniformly over $3 \nmid \xi$; the end of [1, §1.4] states the possible improvement to $O(\exp(-cm))$ and explicitly does not pursue it. A scan of the citing literature (144 citing works at title level, plus arXiv metadata searches for the Syracuse

characteristic function) found no paper supplying the exponential improvement. The mechanism of [1, §7] is a two-dimensional renewal process whose estimate is driven by the number of "white points" along the walk; the m^{-A} loss comes from the largest black triangle, and an exponential bound needs a *linear* lower bound on white points — which is what the structure below supplies, conditionally.

9.2 The proven components

Along a wrapped segment of the walk, the relevant object is the value chain $v(e) = \text{signed}(3^{2j-2}\,2^e \bmod 3^d)$ at effective scale d, with the black set $B_\varepsilon = \{|v| \le \varepsilon 3^d\}$ (Tao's triangles) and its dyadic subsets $D_s = \{v \in B_\varepsilon : \nu_2(v) \ge s\}$.

Lemma 9.1 (doubling without reduction)**.** *If $|v(e)| \le 3^d/4$ then $v(e+1) = 2v(e)$ exactly. Consequently each maximal black run is the doubling chain of a unique odd seed w (the odd part of v), $\{v = 2^t w : \ 0 \le t \le \log_2(\varepsilon 3^d/|w|)\}$, and chains of distinct seeds are disjoint; the seed $w = 3^{2j-2}$ (when below $\varepsilon 3^d$) is the literal small-integer main triangle.*

Lemma 9.2 (pair spacing)**.** *Let w_1, w_2 be the odd seeds of two chains of depth $\ge h$ (i.e. chains reaching $\nu_2 \ge h$, so $|w_i| \le \varepsilon 3^d/2^h$) in the same row, at positions $e_1 < e_2$. Then*

$$e_2 - e_1 \ \ge \ \log_2\!\Big(\frac{2^h}{\varepsilon} - 1\Big) \ \ge \ h + \log_2(1/\varepsilon) - 1.$$

Proof. Put $\Delta = e_2 - e_1 \ge 1$. Then $2^\Delta w_1 \equiv w_2 \pmod{3^d}$ and $2^\Delta w_1 \ne w_2$ (one side is even, the other odd), so $|2^\Delta w_1| + |w_2| \ge 3^d$; with $|w_i| \le \varepsilon 3^d/2^h$ this gives $(2^\Delta + 1)\varepsilon/2^h \ge 1$, which is the claim. □

Lemma 9.3 (2-adic exhaustion, probabilistic form)**.** *Within a black chain, continuation requires $b_j \le \nu_2(v)$ and the budget ν_2 decreases by b at each step; deterministic exit is forced for $b \le \lfloor \log_2(1/2\varepsilon) \rfloor$, while larger b can re-seed only by direct hits of measure $O(\varepsilon 2^b/M)$ per step (measured: 0.2% of transitions). Hence the expected black run per entry is $O(1)$ with geometric tail,* given *the entry-valuation hypothesis below.*

Lemma 9.1 is proved by the displayed inequality ($|2v| \le 3^d/2$ precludes reduction); Lemma 9.3 combines the budget bound with the measured re-seeding rate (the deterministic-exit claim for all b is *false* — an explicit counterexample at $m = 7$, $v = -13$, $b = 7$ exists — which is why the run-length control is routed through the entry-valuation hypothesis).

9.3 The remaining inputs

Hypothesis 9.4 (local entry-valuation bound; "Lemma C-local")**.** *There is an absolute constant C such that, along wrapped segments of length $O(m)$ of the orbit $\{2^e \bmod 3^d\}$ at every effective scale $d \le m$,*

$$\sum_t \mathbb{P}\big(v_{t-1} \notin B,\ v_t \in D_s\big) \ \le \ C\,2^{-s} \sum_t \mathbb{P}\big(v_{t-1} \notin B,\ v_t \in B\big) \qquad (s \ge 0):$$

fresh entries into the ball land on the depth-s subset with at most geometric proportion.

Theorem 6.2 proves exactly this with $C = 1$ when the sums run over *full periods*; the hypothesis is its restriction to windows of length $O(m) = O(\log q)$, $q = 3^d$. Measured behavior: in the deep bands the entry valuations match Geom(1/2) to three digits (0.482/0.272/0.129/0.067); in the shallow shoulder band the tail is enhanced by a bounded factor (≤ 5 at the measured depths), which is harmless since Hypothesis 9.4 permits any absolute C.

Remark 9.5 (what Lemma 9.2 alone gives, and why it is not enough)**.** The pair-spacing lemma does yield an unconditional statement: along a monotone segment of length L the walk enters each chain at most once, an entry of valuation $\geq h$ lies in a chain of depth $\geq h$, and Lemma 9.2 bounds the number of such chains by $L/(h + \log_2(1/\varepsilon) - 1) + 1$; summing over h,

$$\sum_{\text{entries}} \nu_2(v_{\text{entry}}) \;\leq\; L\big(\ln h_{\max} + O(1)\big) + h_{\max}, \qquad h_{\max} \leq \log_2(\varepsilon 3^d),$$

deterministically. This is exactly the shape of the logarithmic loss: the implied bound on black steps, $(L/2)(\ln h_{\max} + O(1))$, exceeds L already for small d, so it guarantees no white points asymptotically, and a per-entry average of order $\log d$ follows only when the number of entries is itself of order L — which is again an equidistribution input. Thus Lemma 9.2 cannot replace Hypothesis 9.4: it controls the *population* of deep chains but not their occupation by the walk, and the geometric depth law is precisely the missing ingredient. The obstruction is not confined to the critical bands: in every band, including the deep band where the product bound contracts at rate ≥ 0.25 numerically, the black points are the small values of the orbit $2^{-n} \bmod 3^d$ over a segment of length $O(m) = O(\log 3^d)$, and bounding their frequency uniformly in m is again Hypothesis 9.4; this is why (BR) is an input in Proposition 9.6 rather than a theorem in any band.

9.4 Classification of the hypothesis

Hypothesis 9.4 is a *finite-orbit, quantitative* form of the Furstenberg $\times 2 \times 3$ phenomenon [9]: it asserts joint 2-adic/3-adic non-concentration for the single multiplicative orbit of 2 modulo powers of 3, in windows of length logarithmic in the modulus. The effective $\times a \times b$ results of [10] concern semigroup orbits and do not reach a single $\times 2$ orbit; the strongest exponential-sum technology for power residues (Postnikov's character formula with Vinogradov/Korobov estimates, and power-modulus refinements) becomes nontrivial only for windows of length $\geq \exp(c(\log q)^{2/3+\epsilon})$, exponentially longer than the $O(\log q)$ required; additive-combinatorial methods require set sizes q^δ. The hypothesis is also adjacent to (but strictly weaker in its averaged form than) the Erdős ternary-digit problem for 2^n [11, 12]. We therefore regard it as precisely isolated but genuinely open; it is, however, decidable for each fixed m on any finite set of frequencies, which is what Section 10 exploits.

9.5 The conditional theorem

Proposition 9.6 (conditional exponential decay)**.** *Assume Hypothesis 9.4, and assume the band-rate estimates*

(BR) for some $\mu < 0$ and all m, the sharpened product bounds along Tao's update process (all-b dynamic program) satisfy $|g_m(2^k)| \leq 2^{-a(\delta)m}$ with $a(\delta) \geq r(\mu) + |\mu|\delta$, $\delta = |k - m\alpha|/m$, for every frequency outside the critical strip $-8 \leq k - m\alpha \leq 2$ of the window, and $a \geq 0.25$ at the seam.

Then $\delta_m \le C\,\mathrm{poly}(m)\,2^{-r(\mu)m}$. *Numerically, (BR) is certified on the extended window with* $\mu = -0.14$ *(*$r = 0.0457$*) at* $m = 192$, 256 *and with* $\mu = -0.15$ *(*$r = 0.0481$*) at* $m = 320$ *(Theorem 10.1); the value* $\mu = -0.16$ *(*$r = 0.0504$*) obtained from rates sampled at* $m = 240$ *is not reached by the certified computations up to* $m = 320$*, and the certified product-bound rates near the window edge extrapolate to* ≈ 0.049 *(Remark 10.2), so we claim only* $r > 0.045$.

Proof architecture. Hypothesis 9.4, with Lemmas 9.1–9.3, gives expected black-run length $O(1)$ per ball entry with geometric tails along every wrapped segment; hence the number of white points is linear in the segment length, at every effective scale, and the octave sum converges (removing the logarithmic divergence responsible for Tao's m^{-A}). This yields the uniform lower bounds on the band rates $a(\delta)$ computed by the all-b product DP — this is where (BR) enters as an input rather than a consequence: the rates sampled at $m = 240$ and 320 (ridge 0.060–0.078 $\le I_0$ as a consistency check, wrap side 0.094–0.17, below-line 0.0633/0.0810/0.108 at $\delta = 0.05/0.15/0.30$, seam ≈ 0.33) close the constraint $r(\mu) + |\mu|\delta \le a(\delta)$ at $\mu = -0.16$, but the certified full-window computations of Section 10 close it only at $\mu = -0.14$ for $m \le 256$, with the shoulder band binding. Given (BR), the capped window functional's one-step contraction at rate $r(\mu)$ is the audited inequality of Computational Fact 4.2, the seam and below-window terms are controlled by (BR), and summing the one-step inequalities gives the claim. □

Remark 9.7. The division of labor is explicit: every step of the architecture is either proved (Theorems 2.1–6.2, Lemmas 9.1–9.2), audited on the window (Computational Fact 4.2), or an input — Hypothesis 9.4 and the band-rate estimates (BR). Theorem 6.2 proves the full-period form of the former; (BR) is certified level by level in Section 10 but not proved uniformly in m. We do not currently see how to prove the windowed form of the hypothesis, and we flag it as the precise obstruction — in our view a more tractable target than the Collatz conjecture itself, but of the $\times 2 \times 3$ family.

10 Certified instances on the window

For each fixed m and each fixed frequency, Hypothesis 9.4 is not needed: Tao's product inequality (7.5) is a theorem, and its right-hand side $B_m(k)$ at the frequency 2^k is a finite computation. We stress at the outset what such computations can and cannot certify: they bound $|g_m(2^k)|$ for the frequencies actually scanned, and no finite scan reaches the $2 \cdot 3^{m-1}$ frequencies of a level $m \ge 16$; statements about δ_m itself therefore rest on the window localization of the extremal frequency (exhaustively verified for $m \le 15$, Computational Fact 7.1(a)).

Theorem 10.1 (finite certification on the window)**.** *Fix* m *and a finite set* W *of exponents. For each* $k \in W$ *the product bound* $B_m(k)$ *is computable with every approximation accounted on the upper side (truncation of the step law, pruning of the dynamic program, and floating-point rounding), yielding numbers* $U_m(k) \ge |g_m(2^k)|$. *On the extended window* $W_m = [-80,\ \lfloor 1.66m \rfloor + 20]$*:*

$m = 64:$	207 *freq.*,	$\sup_{W_m} \lvert g_m \rvert \le 2^{-4.97}$,	$\mu = -0.10$,	$r = 0.0350$,	*slack* $+0.0078$;
$m = 128:$	313 *freq.*,	$\sup_{W_m} \lvert g_m \rvert \le 2^{-8.4}$,	$\mu = -0.12$,	$r = 0.0406$,	*slack* $+0.021$;
$m = 256:$	525 *freq.*,	$\sup_{W_m} \lvert g_m \rvert \le 2^{-15.0}$,	$\mu = -0.14$,	$r = 0.0457$,	*slack* $+0.010$;
$m = 320:$	632 *freq.*,	$\sup_{W_m} \lvert g_m \rvert \le 2^{-18.4}$,	$\mu = -0.15$,	$r = 0.0481$,	*slack* $+0.0035$.

Here the window supremum bound is unconditional, and the last two columns state that the band-rate estimates (BR) hold on W_m *outside the critical strip* $-8 \le u \le 2$ *(which the row-sum*

identity handles) at the given $(\mu, r(\mu))$ with the stated minimal slack (zero safety margin: all budgeted numerical error is inside U_m).

Remark 10.2 (scope)**.** (i) Finer sampling in m shows that the certified constraint rate rises slowly: $\mu = -0.10, -0.11, -0.12, -0.13, -0.14, -0.14, -0.15$ at $m = 64, 96, 128, 160, 192, 256, 320$ ($r = 0.0350, 0.0379, 0.0406, 0.0432, 0.0457, 0.0457, 0.0481$), with the binding constraints in the shoulder band $u \in [-13, -9]$ and the slack shrinking to $+0.0035$ at $m = 320$. The certified window rates fit $a_{\min}(m) = 0.0494 + 0.30 \log_2 m/m$ to within 4×10^{-4} over $m = 64$–320 (shoulder band: $0.048 + 0.40 \log_2 m/m$), so the product-bound rates near the window edge extrapolate to ≈ 0.048–0.049, below $r(-0.16) = 0.0504$; the $(\log_2 m)/m$ correction is the finite-size trace of the logarithmic loss that Hypothesis 9.4 is meant to remove. Whether the sampled rate 0.0504 is attainable by this bookkeeping is therefore open, and we do not claim it. (ii) Under window localization of the extremal frequency, $\delta_m = \sup_{W_m} |g_m|$ and the exact window computation gives the much smaller true value (e.g. $\log_2 \delta_{256} \approx -30.7$); the product bound is the *certifiable* envelope, not the truth. (iii) The error accounting: the step law is truncated at $b \le B(m)$, $B = 41, 65, 112, 135$ for $m = 64, 128, 256, 320$, chosen so that the omitted mass $(m/2) \sum_{b>B} (b-1) 2^{-b}$ (union bound, $|f| \le 1$) is below $2^{-0.35m-8}$; pruned dynamic program states are added back at their true scale (future factors are at most 1), with an m-adaptive threshold $2^{-0.4m-40}$; each $|f|$ receives $+10^{-11}$ and the total a factor $1 + 10^{-8}$. The two allowances are *budgeted*, not interval-certified: they exceed by more than two orders of magnitude the error implied by ≤ 1 ulp accuracy of the library sine and cosine on reduced arguments and by the operation counts of the dynamic program. The certified numbers reproduce the unaccounted floating-point ones to within 10^{-4} in rate.

11 The sharp-rate conjecture and the three clocks

Conjecture 11.1 (sharp rate)**.** $\delta_m = \Theta\big(2^{-I_0 m}\, m^{-c}\big)$ *with* $I_0 = (1 - H(\theta))/\theta = 0.0793186\ldots$ *and some* $c > 0$ *(numerically* $c \approx 1.6$ *for the pinned bridge,* 1.1 *for weighted norms).*

Two asymptotic lemmas would upgrade the numerical support to a proof of the corresponding bounds, and we state them precisely:

$$(\mathrm{L}\alpha) \quad \sup_{k \ge m\alpha + C_0} |g_m(k)| \le 2^{-I_0 m + o(m)} \text{ for some constant } C_0 \qquad (\mathrm{L}\beta) \quad \big|\mathbb{E}_{\mathrm{tilt}}\, \mathrm{e}(\Phi_m)\big| \ \ge\ 1/\mathrm{poly}(m)$$

— above-line control for the upper bound, and pinned-bridge phase non-vanishing (for the bridge pinned at $A_{m-5} = k^*$) for the lower bound; Section 8 locates both. A structural warning is proved along the way: at the extremal frequency the ridge phases are $1 + O(\rho^m)$, so no finite-depth expansion of the recursion can prove either lemma — the decay of δ_m is an unbounded-memory effect, the Fourier-side analogue of the finite-window obstructions of [15].

Three clocks

The constant I_0 is one number read in three clocks: (a) the stopping-time/certificate density exponent $1 - H(\theta)$ per binary digit [3, 15]; (b) the KL anomaly rate $D(\mathrm{Geom}(\theta) \| \mathrm{Geom}(1/2))$ per odd step, forced on every non-descending window by the deterministic rigidity identity; (c) the 3-adic mixing rate per ternary level (Conjecture 11.1). Equalities (a)=(b) are one-line clock conversions ((3)); the substantive claim is that the Fourier clock (c) shows the same time, and

the mechanism behind it is the localization of the extremal frequency on the cycle-resonance line $2^{k^*} \approx 3^m$ (Computational Fact 7.1), where the large-deviation cost of tracking the line is exactly I_0.

12 Relation to Tao's program and to the certificate barrier

Tao's transport argument consumes the 3-adic equidistribution of an *ensemble* of initial points; its pointwise failure is structural (finite-window realization obstructions [15]). What an exponential δ_m would buy is the *effective constant* in that program: the number of ternary levels needed per unit of density gain is $1/I_0 \approx 12.6$, and the boundary-layer analysis shows this constant is sharp at the level of the Fourier input. Conversely, the barrier theory of [15] prices the 2-adic certificate side of the same constant, and the extremal objects on both sides — the all-ones tower $-1 \bmod 2^m$ there, the pinned bridge on $2^k \approx 3^m$ here — are the two faces of the cycle-resonance line. We find it notable that the obstruction isolated in Hypothesis 9.4 is again a $\times 2 \times 3$ statement: all roads from this problem's analytic side appear to lead to the same rigidity frontier.

Since [1] appeared, manuscripts not yet peer-reviewed have claimed the upgrade of its logarithmic-density conclusion to natural density ([1, Remark 1.16]) through sum-conditioned versions of the superpolynomial decay [2]. Whatever their eventual status, they use only superpolynomial decay of δ_m; the exponential question studied here is independent of them.

Funding

No funding was obtained for this work.

Disclosure statement

The author reports there are no competing interests to declare.

Declaration of generative AI use

Generative AI (Anthropic's Claude) was used, under the author's direction, to assist in drafting and revising the manuscript, in developing and checking the computational scripts and the Lean 4 proof files, and in searching the literature. All definitions, theorems, and proofs were reviewed by the author, the computational claims are backed by the verifiable artifacts described in this article, and the author takes full responsibility for its content.

Data availability statement

All code and data supporting this study — the exponent-lattice computations, the audits, the band-rate dynamic programs, the certification drivers with their logs, and one self-contained script per computational claim, with a claim-to-script map and SHA-256 manifest — are openly available in the reproduction repository `collatz-barrier-repro` (https://github.com/WakauraH/collatz-barrier-repro); a DOI-stamped archival snapshot of the exact version used here will

be deposited upon acceptance. The same material is included as the supplementary bundle of this submission.

References

[1] T. Tao, *Almost all orbits of the Collatz map attain almost bounded values*, Forum Math. Pi **10** (2022), e12.

[2] J. Allikvere, *Almost all Collatz orbits attain almost bounded values in natural density*, v2, Zenodo (July 2026), https://doi.org/10.5281/zenodo.21499244 (not yet peer-reviewed).

[3] R. Terras, *A stopping time problem on the positive integers*, Acta Arith. **30** (1976), 241–252.

[4] J. C. Lagarias, *The* $3x+1$ *problem and its generalizations*, Amer. Math. Monthly **92** (1985), 3–23.

[5] J. C. Lagarias (ed.), *The Ultimate Challenge: The* $3x+1$ *Problem*, American Mathematical Society, Providence, RI, 2010.

[6] A. V. Kontorovich and J. C. Lagarias, *Stochastic models for the* $3x+1$ *and* $5x+1$ *problems and related problems*, in: The Ultimate Challenge: The $3x+1$ Problem, Amer. Math. Soc., 2010, 131–188.

[7] H. Kesten, *Random difference equations and renewal theory for products of random matrices*, Acta Math. **131** (1973), 207–248.

[8] C. M. Goldie, *Implicit renewal theory and tails of solutions of random equations*, Ann. Appl. Probab. **1** (1991), 126–166.

[9] H. Furstenberg, *Disjointness in ergodic theory, minimal sets, and a problem in Diophantine approximation*, Math. Systems Theory **1** (1967), 1–49.

[10] J. Bourgain, E. Lindenstrauss, P. Michel, and A. Venkatesh, *Some effective results for* $\times a \times b$, Ergodic Theory Dynam. Systems **29** (2009), 1705–1722.

[11] P. Erdős, *Some unconventional problems in number theory*, Math. Mag. **52** (1979), 67–70.

[12] J. C. Lagarias, *Ternary expansions of powers of 2*, J. Lond. Math. Soc. (2) **79** (2009), 562–588.

[13] D. Barina, *Convergence verification of the Collatz problem*, J. Supercomput. **77** (2021), 2681–2688.

[14] G. J. Wirsching, *The Dynamical System Generated by the* $3n+1$ *Function*, Lecture Notes in Math. **1681**, Springer, Berlin, 1998.

[15] H. Wakaura, *Certificate complexity of uniform descent for the* $3x+1$ *problem*, preprint (2026).

A Data, scripts, and pre-registration log

Scripts

Each computational claim maps to one self-contained exact-arithmetic script in the supplementary bundle: the exponent-lattice recursion and full scans; the rescaled window computation to $m = 1200$; the first-passage decomposition audit (identity to 10^{-14}); the weighted-norm and capped-ℓ^1 audits (399 levels, zero violations); the crossing/corner/pinned-bridge decompositions; the ℓ^4 collision functional; the all-b product DP for band rates; the cycle-law verification of Theorem 6.2; and the certification driver of Theorem 10.1 for $m = 64, 128, 256$.

Pre-registration log (calibration)

Major numerical discriminations were decided by criteria declared before the deciding computation, and we report all outcomes, including negative ones, as a calibration record. (1) An initial candidate for the decay constant, $\rho = 2^{3/2}/3$ (suggested by an apparent plateau at $m \le 60$), was rejected by the pre-registered discrimination test at $m \le 200$: the extremal exponent ratio k^*/m crossed $3/2$ and continued toward $\log_2 3$, and the local rate descended to the I_0 prediction; the plateau was a finite-size crossing effect. (2) A pre-registered main-term (positivity-based) lower-bound construction was rejected by its own criterion (measured rate $I_0 + 0.0093$, below target), leading to the crossing-term mechanism of Section 8. (3) The pre-registered two-sided band $[0.0790, 0.0797]$ for the asymptotic rate was *not* met (the measured range is $[0.0790, 0.0800]$). After seeing the data we adopted a one-sided criterion (rate $\ge I_0 - \epsilon$, approaching I_0), on the structural ground that certified functionals bound the rate from one side only; the data satisfy that criterion, but this is a post-hoc replacement and we record it as such. We keep this log because, in our experience, the two rejections were the direct source of the mechanism results in Section 8.